NEETCODE 150 CURRICULUM

Swift 6 Master Manual
Easy-to-Hard Sequenced Master Guide (#1 to #150)

Complete Swift 6 Code Solutions with Xcode Syntax Highlighting & Clickable LeetCode Problem URLs

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Volume: Easy-to-Hard Sequenced Master Guide (#1 to #150) **Problems Count:** 150 Solved Problems

Language: Swift 6 Idiomatic Code **Hyperlinks:** 100% Direct LeetCode URLs

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#	Problem Title	Category	Difficulty
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■ EASY LEVEL PROBLEMS

• Arrays & Hashing

#1. Contains Duplicate (Easy) [■ Open LeetCode Problem]

Logic: *Use Set to store seen numbers.*

```
class Solution {
    func containsDuplicate(_ nums: [Int]) -> Bool {
        return Set(nums).count < nums.count
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#2. Valid Anagram (Easy) [■ Open LeetCode Problem]

Logic: *Compare sorted character arrays.*

```
class Solution {
    func isAnagram(_ s: String, _ t: String) -> Bool {
        return s.sorted() == t.sorted()
    }
}
```

Time Complexity: O(N log N) **Space Complexity:** O(N)

#3. Two Sum (Easy) [■ Open LeetCode Problem]

Logic: *Hash Map storing number -> index.*

```
class Solution {
    func twoSum(_ nums: [Int], _ target: Int) -> [Int] {
        var map = [Int: Int]()
        for (i, n) in nums.enumerated() {
            if let idx = map[target - n] { return [idx, i] }
            map[n] = i
        }
        return []
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Two Pointers

#4. Valid Palindrome (Easy) [■ Open LeetCode Problem]

Logic: *Filter alphanumeric and two pointer scan.*

```
class Solution {
    func isPalindrome(_ s: String) -> Bool {
        let c = Array(s.lowercased().filter { $0.isLetter || $0.isNumber })
        var l = 0, r = c.count - 1
        while l < r {
            if c[l] != c[r] { return false }
            l += 1; r -= 1
        }
        return true
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Sliding Window

#5. Best Time to Buy and Sell Stock (Easy) [■ Open LeetCode Problem]

Logic: *Track minPrice and maxProfit.*

```
class Solution {
    func maxProfit(_ prices: [Int]) -> Int {
        var minP = Int.max, maxP = 0
        for p in prices { minP = min(minP, p); maxP = max(maxP, p - minP) }
        return maxP
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(1)

• Stack

#6. Valid Parentheses (Easy) [■ Open LeetCode Problem]

Logic: *Stack push open brackets, pop and match closing.*

```
class Solution {
    func isValid(_ s: String) -> Bool {
        var st = [Character]()
        for c in s {
            if c == "(" { st.append(")") }
            else if c == "[" { st.append("]") }
            else if c == "{" { st.append("}") }
            else if st.isEmpty || st.removeLast() != c { return false }
        }
        return st.isEmpty
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Binary Search

#7. Binary Search (Easy) [■ Open LeetCode Problem]

Logic: *Iterative mid calculation.*

```
class Solution {
    func search(_ nums: [Int], _ target: Int) -> Int {
        var l = 0, r = nums.count - 1
        while l <= r {
            let m = l + (r - l) / 2
            if nums[m] == target { return m }
            else if nums[m] < target { l = m + 1 } else { r = m - 1 }
        }
        return -1
    }
}
```

Time Complexity: O(log N) **Space Complexity:** O(1)

• Graphs

#8. Reverse Linked List #35 (Easy) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Reverse Linked List.*

```
class Solution_35 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Reverse Linked List
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

■ MEDIUM LEVEL PROBLEMS

• Arrays & Hashing

#9. Group Anagrams (Medium) [■ Open LeetCode Problem]

Logic: *Sorted string key mapping in Hash Map.*

```
class Solution {
    func groupAnagrams(_ strs: [String]) -> [[String]] {
        var map = [String: [String]]()
        for s in strs { map[String(s.sorted()), default: []].append(s) }
        return Array(map.values)
    }
}
```

Time Complexity: O(N * K log K) **Space Complexity:** O(N * K)

#10. Top K Frequent Elements (Medium) [■ Open LeetCode Problem]

Logic: *Count frequencies and bucket sort.*

```
class Solution {
    func topKFrequent(_ nums: [Int], _ k: Int) -> [Int] {
        var counts = [Int: Int]()
        for n in nums { counts[n, default: 0] += 1 }
        return Array(counts.keys.sorted{ counts[$0]! > counts[$1]! }.prefix(k))
    }
}
```

Time Complexity: O(N log N) **Space Complexity:** O(N)

#11. Product of Array Except Self (Medium) [■ Open LeetCode Problem]

Logic: *Prefix and suffix product passes.*

```
class Solution {
    func productExceptSelf(_ nums: [Int]) -> [Int] {
        var res = Array(repeating: 1, count: nums.count), p = 1
        for i in 0..<nums.count { res[i] = p; p *= nums[i] }
        p = 1
        for i in (0..<nums.count).reversed() { res[i] *= p; p *= nums[i] }
        return res
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(1)

#12. Valid Sudoku (Medium) [■ Open LeetCode Problem]

Logic: *Hash Sets for rows, columns, and 3x3 boxes.*

```
class Solution {
    func isValidSudoku(_ board: [[Character]]) -> Bool {
        var r = Array(repeating: Set<Character>(), count: 9), c = r, b = r
        for i in 0..<9 {
            for j in 0..<9 {
                let char = board[i][j]
                if char == "." { continue }
                let idx = (i / 3) * 3 + (j / 3)
                if r[i].contains(char) || c[j].contains(char) || b[idx].contains(char) { return false }
                r[i].insert(char); c[j].insert(char); b[idx].insert(char)
            }
        }
        return true
    }
}
```

Time Complexity: O(1) **Space Complexity:** O(1)

#13. Encode and Decode Strings (Medium) [■ Open LeetCode Problem]

Logic: *Length prefix encoding format len#str.*

```
class Solution {
    func encode(_ strs: [String]) -> String {
        return strs.map { "\($0.count)#\($0)" }.joined()
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#14. Longest Consecutive Sequence (Medium) [■ Open LeetCode Problem]

Logic: *Set lookup start sequence !set.contains(n-1).*

```
class Solution {
    func longestConsecutive(_ nums: [Int]) -> Int {
        let set = Set(nums); var maxL = 0
        for n in set {
            if !set.contains(n - 1) {
                var curr = n, len = 1
                while set.contains(curr + 1) { curr += 1; len += 1 }
                maxL = max(maxL, len)
            }
        }
        return maxL
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Two Pointers

#15. Two Sum II Input Array Is Sorted (Medium) [■ Open LeetCode Problem]

Logic: *Two pointers left and right on sorted array.*

```
class Solution {
    func twoSum(_ numbers: [Int], _ target: Int) -> [Int] {
        var l = 0, r = numbers.count - 1
        while l < r {
            let s = numbers[l] + numbers[r]
            if s == target { return [l + 1, r + 1] }
            else if s < target { l += 1 }
            else { r -= 1 }
        }
        return []
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(1)

#16. 3Sum (Medium) [■ Open LeetCode Problem]

Logic: *Sort array, fix i, two pointers l and r.*

```
class Solution {
    func threeSum(_ nums: [Int]) -> [[Int]] {
        let s = nums.sorted(); var res = [[Int]]()
        for i in 0..<s.count {
            if i > 0 && s[i] == s[i-1] { continue }
            var l = i + 1, r = s.count - 1
            while l < r {
                let sum = s[i] + s[l] + s[r]
                if sum == 0 {
                    res.append([s[i], s[l], s[r]])
                    while l < r && s[l] == s[l+1] { l += 1 }
                    while l < r && s[r] == s[r-1] { r -= 1 }
                    l += 1; r -= 1
                } else if sum < 0 { l += 1 } else { r -= 1 }
            }
        }
        return res
    }
}
```

Time Complexity: O(N^2) **Space Complexity:** O(1)

#17. Container With Most Water (Medium) [■ Open LeetCode Problem]

Logic: *Two pointers at bounds, shrink smaller side.*

```
class Solution {
    func maxArea(_ height: [Int]) -> Int {
        var l = 0, r = height.count - 1, maxA = 0
        while l < r {
            maxA = max(maxA, min(height[l], height[r]) * (r - l))
            if height[l] < height[r] { l += 1 } else { r -= 1 }
        }
        return maxA
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(1)

• Sliding Window

#18. Longest Substring Without Repeating Characters (Medium) [■ Open LeetCode Problem]

Logic: *Sliding window with map of last seen char index.*

```
class Solution {
    func lengthOfLongestSubstring(_ s: String) -> Int {
        var map = [Character: Int](), l = 0, maxL = 0
        for (r, c) in s.enumerated() {
            if let pos = map[c], pos >= l { l = pos + 1 }
            map[c] = r
            maxL = max(maxL, r - l + 1)
        }
        return maxL
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#19. Longest Repeating Character Replacement (Medium) [■ Open LeetCode Problem]

Logic: *Sliding window windowLen - maxFreq <= k.*

```
class Solution {
    func characterReplacement(_ s: String, _ k: Int) -> Int {
        var counts = [Character: Int](), l = 0, maxF = 0, maxL = 0, chars = Array(s)
        for r in 0..<chars.count {
            counts[chars[r], default: 0] += 1
            maxF = max(maxF, counts[chars[r]]!)
            if (r - l + 1) - maxF > k { counts[chars[l]]! -= 1; l += 1 }
            maxL = max(maxL, r - l + 1)
        }
        return maxL
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(26)

#20. Permutation in String (Medium) [■ Open LeetCode Problem]

Logic: *Fixed window size s1.count matching freq arrays.*

```
class Solution {
    func checkInclusion(_ s1: String, _ s2: String) -> Bool {
        if s1.count > s2.count { return false }
        var c1 = Array(repeating: 0, count: 26), c2 = c1, a = Int(Character("a").asciiValue!)
        for c in s1 { c1[Int(c.asciiValue!) - a] += 1 }
        let chars = Array(s2)
        for i in 0..<chars.count {
            c2[Int(chars[i].asciiValue!) - a] += 1
            if i >= s1.count { c2[Int(chars[i - s1.count].asciiValue!) - a] -= 1 }
            if c1 == c2 { return true }
        }
        return false
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(26)

• Stack

#21. Min Stack (Medium) [■ Open LeetCode Problem]

Logic: *Dual stacks mainStack and minStack.*

```
class MinStack {
    var main = [Int](), minS = [Int]()
    func push(_ val: Int) { main.append(val); minS.append(min(val, minS.last ?? Int.max)) }
    func pop() { main.removeLast(); minS.removeLast() }
    func top() -> Int { main.last! }
    func getMin() -> Int { minS.last! }
}
```

Time Complexity: O(1) **Space Complexity:** O(N)

#22. Evaluate Reverse Polish Notation (Medium) [■ Open LeetCode Problem]

Logic: *Stack store operands, pop two on operator.*

```
class Solution {
    func evalRPN(_ tokens: [String]) -> Int {
        var st = [Int]()
        for t in tokens {
            if let val = Int(t) { st.append(val) }
            else {
                let b = st.removeLast(), a = st.removeLast()
                switch t {
                case "+": st.append(a + b)
                case "-": st.append(a - b)
                case "*": st.append(a * b)
                default: st.append(a / b)
                }
            }
        }
        return st.last!
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#23. Generate Parentheses (Medium) [■ Open LeetCode Problem]

Logic: *Backtracking open < n and close < open.*

```
class Solution {
    func generateParenthesis(_ n: Int) -> [String] {
        var res = [String]()
        func backtrack(_ s: String, _ open: Int, _ close: Int) {
            if s.count == 2 * n { res.append(s); return }
            if open < n { backtrack(s + "(", open + 1, close) }
            if close < open { backtrack(s + ")", open, close + 1) }
        }
        backtrack("", 0, 0)
        return res
    }
}
```

Time Complexity: O(4^N / sqrt(N)) **Space Complexity:** O(N)

#24. Daily Temperatures (Medium) [■ Open LeetCode Problem]

Logic: *Monotonic decreasing stack storing indices.*

```
class Solution {
    func dailyTemperatures(_ temp: [Int]) -> [Int] {
        var res = Array(repeating: 0, count: temp.count), st = [Int]()
        for i in 0..<temp.count {
            while !st.isEmpty && temp[i] > temp[st.last!] {
                let prev = st.removeLast()
                res[prev] = i - prev
            }
            st.append(i)
        }
        return res
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#25. Car Fleet (Medium) [■ Open LeetCode Problem]

Logic: *Sort by position desc, compute target arrival time.*

```
class Solution {
    func carFleet(_ target: Int, _ position: [Int], _ speed: [Int]) -> Int {
        let cars = zip(position, speed).sorted { $0.0 > $1.0 }
        var st = [Double]()
        for (p, s) in cars {
            let time = Double(target - p) / Double(s)
            if st.isEmpty || time > st.last! { st.append(time) }
        }
        return st.count
    }
}
```

Time Complexity: O(N log N) **Space Complexity:** O(N)

• Binary Search

#26. Search a 2D Matrix (Medium) [■ Open LeetCode Problem]

Logic: *Virtual 1D binary search over matrix.*

```swift
class Solution {
    func searchMatrix(_ matrix: [[Int]], _ target: Int) -> Bool {
        let R = matrix.count, C = matrix[0].count
        var l = 0, r = R * C - 1
        while l <= r {
            let m = l + (r - l) / 2, val = matrix[m / C][m % C]
            if val == target { return true }
            else if val < target { l = m + 1 } else { r = m - 1 }
        }
        return false
    }
}
```

Time Complexity: O(log(M*N)) **Space Complexity:** O(1)

#27. Koko Eating Bananas (Medium) [■ Open LeetCode Problem]

Logic: *Binary search on eating speed k.*

```swift
class Solution {
    func minEatingSpeed(_ piles: [Int], _ h: Int) -> Int {
        var l = 1, r = piles.max()!, ans = r
        while l <= r {
            let k = l + (r - l) / 2, hrs = piles.reduce(0) { $0 + ($1 + k - 1) / k }
            if hrs <= h { ans = k; r = k - 1 } else { l = k + 1 }
        }
        return ans
    }
}
```

Time Complexity: O(N log M) **Space Complexity:** O(1)

#28. Find Minimum in Rotated Sorted Array (Medium) [■ Open LeetCode Problem]

Logic: *Compare mid with right boundary.*

```swift
class Solution {
    func findMin(_ nums: [Int]) -> Int {
        var l = 0, r = nums.count - 1
        while l < r {
            let m = l + (r - l) / 2
            if nums[m] > nums[r] { l = m + 1 } else { r = m }
        }
        return nums[l]
    }
}
```

Time Complexity: O(log N) **Space Complexity:** O(1)

#29. Search in Rotated Sorted Array (Medium) [■ Open LeetCode Problem]

Logic: *Determine sorted half at mid.*

```swift
class Solution {
    func search(_ nums: [Int], _ target: Int) -> Int {
        var l = 0, r = nums.count - 1
        while l <= r {
            let m = l + (r - l) / 2
            if nums[m] == target { return m }
            if nums[l] <= nums[m] {
                if nums[l] <= target && target < nums[m] { r = m - 1 }
                else { l = m + 1 }
            } else {
                if nums[m] < target && target <= nums[r] { l = m + 1 }
                else { r = m - 1 }
            }
        }
        return -1
    }
}
```

Time Complexity: O(log N) **Space Complexity:** O(1)

#30. Time Based Key-Value Store (Medium) [■ Open LeetCode Problem]

Logic: *Binary search on list of (timestamp, value).*

```swift
class TimeMap {
    var map = [String: [(Int, String)]]()
    func set(_ key: String, _ value: String, _ timestamp: Int) {
        map[key, default: []].append((timestamp, value))
    }
    func get(_ key: String, _ timestamp: Int) -> String {
        guard let list = map[key] else { return "" }
        var l = 0, r = list.count - 1, ans = ""
        while l <= r {
            let m = l + (r - l) / 2
            if list[m].0 <= timestamp { ans = list[m].1; l = m + 1 }
            else { r = m - 1 }
        }
        return ans
    }
}
```

Time Complexity: O(log N) **Space Complexity:** O(N)

• Dynamic Programming

#31. Merge Two Sorted Lists #36 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Merge Two Sorted Lists.*

```swift
class Solution_36 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Merge Two Sorted Lists
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#32. Reorder List #37 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Reorder List.*

```
class Solution_37 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Reorder List
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#33. Remove Nth Node From End of List #38 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Remove Nth Node From End of List.*

```
class Solution_38 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Remove Nth Node From End of List
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#34. Copy List with Random Pointer #39 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Copy List with Random Pointer.*

```
class Solution_39 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Copy List with Random Pointer
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#35. Add Two Numbers #40 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Add Two Numbers.*

```
class Solution_40 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Add Two Numbers
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#36. Linked List Cycle #41 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Linked List Cycle.*

```
class Solution_41 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Linked List Cycle
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#37. Find the Duplicate Number #42 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Find the Duplicate Number.*

```
class Solution_42 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Find the Duplicate Number
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#38. LRU Cache #43 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for LRU Cache.*

```
class Solution_43 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for LRU Cache
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#39. Merge K Sorted Lists #44 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Merge K Sorted Lists.*

```
class Solution_44 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Merge K Sorted Lists
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#40. Reverse Nodes in k-Group #45 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Reverse Nodes in k-Group.*

```
class Solution_45 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Reverse Nodes in k-Group
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#41. Invert Binary Tree #46 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Invert Binary Tree.*

```
class Solution_46 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Invert Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#42. Maximum Depth of Binary Tree #47 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Maximum Depth of Binary Tree.*

```
class Solution_47 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Maximum Depth of Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#43. Diameter of Binary Tree #48 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Diameter of Binary Tree.*

```
class Solution_48 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Diameter of Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#44. Balanced Binary Tree #49 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Balanced Binary Tree.*

```
class Solution_49 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Balanced Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#45. Same Tree #50 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Same Tree.*

```
class Solution_50 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Same Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#46. Subtree of Another Tree #51 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Subtree of Another Tree.*

```
class Solution_51 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Subtree of Another Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#47. Lowest Common Ancestor of a BST #52 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Lowest Common Ancestor of a BST.*

```
class Solution_52 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Lowest Common Ancestor of a BST
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#48. Binary Tree Level Order Traversal #53 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Binary Tree Level Order Traversal.*

```
class Solution_53 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Binary Tree Level Order Traversal
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#49. Binary Tree Right Side View #54 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Binary Tree Right Side View.*

```
class Solution_54 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Binary Tree Right Side View
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#50. Count Good Nodes in Binary Tree #55 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Count Good Nodes in Binary Tree.*

```
class Solution_55 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Count Good Nodes in Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#51. Validate Binary Search Tree #56 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Validate Binary Search Tree.*

```
class Solution_56 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Validate Binary Search Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#52. Kth Smallest Element in a BST #57 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Kth Smallest Element in a BST.*

```
class Solution_57 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Kth Smallest Element in a BST
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#53. Construct Binary Tree from Preorder and Inorder Traversal #58 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Construct Binary Tree from Preorder and Inorder Traversal.*

```
class Solution_58 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Construct Binary Tree from Preorder and Inorder Traversal
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#54. Binary Tree Maximum Path Sum #59 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Binary Tree Maximum Path Sum.*

```
class Solution_59 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Binary Tree Maximum Path Sum
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#55. Serialize and Deserialize Binary Tree #60 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Serialize and Deserialize Binary Tree.*

```
class Solution_60 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Serialize and Deserialize Binary Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#56. Implement Trie (Prefix Tree) #61 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Implement Trie (Prefix Tree).*

```
class Solution_61 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Implement Trie (Prefix Tree)
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#57. Design Add and Search Words Data Structure #62 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Design Add and Search Words Data Structure.*

```
class Solution_62 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Design Add and Search Words Data Structure
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#58. Word Search II #63 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Word Search II.*

```
class Solution_63 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Word Search II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#59. Kth Largest Element in a Stream #64 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Kth Largest Element in a Stream.*

```
class Solution_64 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Kth Largest Element in a Stream
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#60. Last Stone Weight #65 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Last Stone Weight.*

```
class Solution_65 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Last Stone Weight
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#61. K Closest Points to Origin #66 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for K Closest Points to Origin.*

```
class Solution_66 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for K Closest Points to Origin
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#62. Kth Largest Element in an Array #67 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Kth Largest Element in an Array.*

```
class Solution_67 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Kth Largest Element in an Array
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#63. Task Scheduler #68 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Task Scheduler.*

```
class Solution_68 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Task Scheduler
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#64. Design Twitter #69 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Design Twitter.*

```
class Solution_69 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Design Twitter
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#65. Find Median from Data Stream #70 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Find Median from Data Stream.*

```
class Solution_70 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Find Median from Data Stream
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#66. Subsets #71 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Subsets.*

```
class Solution_71 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Subsets
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#67. Combination Sum #72 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Combination Sum.*

```
class Solution_72 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Combination Sum
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#68. Permutations #73 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Permutations.*

```swift
class Solution_73 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Permutations
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#69. Subsets II #74 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Subsets II.*

```swift
class Solution_74 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Subsets II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#70. Combination Sum II #75 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Combination Sum II.*

```swift
class Solution_75 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Combination Sum II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#71. Word Search #76 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Word Search.*

```swift
class Solution_76 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Word Search
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#72. Palindrome Partitioning #77 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Palindrome Partitioning.*

```
class Solution_77 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Palindrome Partitioning
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#73. Letter Combinations of a Phone Number #78 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Letter Combinations of a Phone Number.*

```
class Solution_78 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Letter Combinations of a Phone Number
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#74. N-Queens #79 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for N-Queens.*

```
class Solution_79 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for N-Queens
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#75. Number of Islands #80 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Number of Islands.*

```
class Solution_80 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Number of Islands
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#76. Max Area of Island #81 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Max Area of Island.*

```
class Solution_81 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Max Area of Island
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#77. Clone Graph #82 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Clone Graph.*

```
class Solution_82 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Clone Graph
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#78. Walls and Gates #83 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Walls and Gates.*

```
class Solution_83 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Walls and Gates
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#79. Rotting Oranges #84 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Rotting Oranges.*

```
class Solution_84 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Rotting Oranges
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#80. Pacific Atlantic Water Flow #85 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Pacific Atlantic Water Flow.*

```
class Solution_85 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Pacific Atlantic Water Flow
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#81. Surrounded Regions #86 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Surrounded Regions.*

```
class Solution_86 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Surrounded Regions
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#82. Course Schedule #87 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Course Schedule.*

```
class Solution_87 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Course Schedule
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#83. Course Schedule II #88 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Course Schedule II.*

```
class Solution_88 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Course Schedule II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#84. Graph Valid Tree #89 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Graph Valid Tree.*

```
class Solution_89 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Graph Valid Tree
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#85. Number of Connected Components #90 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Number of Connected Components.*

```
class Solution_90 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Number of Connected Components
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#86. Redundant Connection #91 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Redundant Connection.*

```
class Solution_91 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Redundant Connection
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#87. Word Ladder #92 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Word Ladder.*

```
class Solution_92 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Word Ladder
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#88. Reconstruct Itinerary #93 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Reconstruct Itinerary.*

```swift
class Solution_93 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Reconstruct Itinerary
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#89. Min Cost to Connect All Points #94 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Min Cost to Connect All Points.*

```swift
class Solution_94 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Min Cost to Connect All Points
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#90. Swim in Rising Water #95 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Swim in Rising Water.*

```swift
class Solution_95 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Swim in Rising Water
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#91. Alien Dictionary #96 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Alien Dictionary.*

```swift
class Solution_96 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Alien Dictionary
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#92. Cheapest Flights Within K Stops #97 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Cheapest Flights Within K Stops.*

```
class Solution_97 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Cheapest Flights Within K Stops
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#93. Network Delay Time #98 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Network Delay Time.*

```
class Solution_98 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Network Delay Time
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#94. Climbing Stairs #99 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Climbing Stairs.*

```
class Solution_99 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Climbing Stairs
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#95. Min Cost Climbing Stairs #100 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Min Cost Climbing Stairs.*

```
class Solution_100 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Min Cost Climbing Stairs
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#96. House Robber #101 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for House Robber.*

```
class Solution_101 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for House Robber
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#97. House Robber II #102 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for House Robber II.*

```
class Solution_102 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for House Robber II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#98. Longest Palindromic Substring #103 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Longest Palindromic Substring.*

```
class Solution_103 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Longest Palindromic Substring
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#99. Palindromic Substrings #104 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Palindromic Substrings.*

```
class Solution_104 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Palindromic Substrings
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#100. Decode Ways #105 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Decode Ways.*

```
class Solution_105 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Decode Ways
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#101. Coin Change #106 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Coin Change.*

```
class Solution_106 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Coin Change
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#102. Maximum Product Subarray #107 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Maximum Product Subarray.*

```
class Solution_107 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Maximum Product Subarray
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#103. Word Break #108 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Word Break.*

```
class Solution_108 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Word Break
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#104. Longest Increasing Subsequence #109 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Longest Increasing Subsequence.*

```
class Solution_109 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Longest Increasing Subsequence
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#105. Partition Equal Subset Sum #110 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Partition Equal Subset Sum.*

```
class Solution_110 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Partition Equal Subset Sum
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#106. Unique Paths #111 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Unique Paths.*

```
class Solution_111 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Unique Paths
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#107. Longest Common Subsequence #112 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Longest Common Subsequence.*

```
class Solution_112 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Longest Common Subsequence
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#108. Stock with Cooldown #113 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Stock with Cooldown.*

```
class Solution_113 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Stock with Cooldown
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#109. Coin Change II #114 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Coin Change II.*

```
class Solution_114 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Coin Change II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#110. Target Sum #115 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Target Sum.*

```
class Solution_115 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Target Sum
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#111. Interleaving String #116 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Interleaving String.*

```
class Solution_116 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Interleaving String
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#112. Longest Increasing Path in a Matrix #117 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Longest Increasing Path in a Matrix.*

```
class Solution_117 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Longest Increasing Path in a Matrix
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#113. Distinct Subsequences #118 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Distinct Subsequences.*

```
class Solution_118 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Distinct Subsequences
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#114. Edit Distance #119 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Edit Distance.*

```
class Solution_119 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Edit Distance
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#115. Burst Balloons #120 (Medium) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Burst Balloons.*

```
class Solution_120 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Burst Balloons
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

■ HARD LEVEL PROBLEMS

• Two Pointers

#116. Trapping Rain Water (Hard) [■ Open LeetCode Problem]

Logic: *Two pointers with maxLeft and maxRight.*

```
class Solution {
    func trap(_ height: [Int]) -> Int {
        var l = 0, r = height.count - 1, maxL = 0, maxR = 0, ans = 0
        while l < r {
            if height[l] < height[r] {
                if height[l] >= maxL { maxL = height[l] } else { ans += maxL - height[l] }
                l += 1
            } else {
                if height[r] >= maxR { maxR = height[r] } else { ans += maxR - height[r] }
                r -= 1
            }
        }
        return ans
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(1)

• Sliding Window

#117. Minimum Window Substring (Hard) [■ Open LeetCode Problem]

Logic: *Sliding window frequency count match.*

```
class Solution {
    func minWindow(_ s: String, _ t: String) -> String {
        var target = [Character: Int]()
        for c in t { target[c, default: 0] += 1 }
        var required = target.count, formed = 0, window = [Character: Int](), l = 0, r = 0, ans = (-1, 0, 0),
sArr = Array(s)
        while r < sArr.count {
            let c = sArr[r]
            window[c, default: 0] += 1
            if let count = target[c], window[c] == count { formed += 1 }
            while l <= r && formed == required {
                if ans.0 == -1 || r - l + 1 < ans.0 { ans = (r - l + 1, l, r) }
                let lc = sArr[l]
                window[lc]! -= 1
                if let count = target[lc], window[lc]! < count { formed -= 1 }
                l += 1
            }
            r += 1
        }
        return ans.0 == -1 ? "" : String(sArr[ans.1...ans.2])
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

#118. Sliding Window Maximum (Hard) [■ Open LeetCode Problem]

Logic: *Monotonic decreasing Deque storing indices.*

```swift
class Solution {
    func maxSlidingWindow(_ nums: [Int], _ k: Int) -> [Int] {
        var q = [Int](), res = [Int]()
        for i in 0..<nums.count {
            while !q.isEmpty && nums[q.last!] < nums[i] { q.removeLast() }
            q.append(i)
            if q.first! == i - k { q.removeFirst() }
            if i >= k - 1 { res.append(nums[q.first!]) }
        }
        return res
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(K)

• Stack

#119. Largest Rectangle in Histogram (Hard) [■ Open LeetCode Problem]

Logic: *Monotonic increasing stack storing (index, height).*

```swift
class Solution {
    func largestRectangleArea(_ heights: [Int]) -> Int {
        var st = [(Int, Int)](), maxA = 0, h = heights + [0]
        for (i, height) in h.enumerated() {
            var start = i
            while !st.isEmpty && st.last!.1 > height {
                let (pIdx, pHeight) = st.removeLast()
                maxA = max(maxA, pHeight * (i - pIdx))
                start = pIdx
            }
            st.append((start, height))
        }
        return maxA
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Binary Search

#120. Median of Two Sorted Arrays (Hard) [■ Open LeetCode Problem]

Logic: *Binary search partition on smaller array.*

```
class Solution {
    func findMedianSortedArrays(_ A: [Int], _ B: [Int]) -> Double {
        let (a, b) = A.count <= B.count ? (A, B) : (B, A), m = a.count, n = b.count
        var l = 0, r = m
        while l <= r {
            let i = l + (r - l) / 2, j = (m + n + 1) / 2 - i
            let maxL1 = i == 0 ? Int.min : a[i - 1], minR1 = i == m ? Int.max : a[i]
            let maxL2 = j == 0 ? Int.min : b[j - 1], minR2 = j == n ? Int.max : b[j]
            if maxL1 <= minR1 && maxL2 <= minR1 && maxL1 <= minR2 && maxL2 <= minR2 {
                if (m + n) % 2 == 1 { return Double(max(maxL1, maxL2)) }
                else { return Double(max(maxL1, maxL2) + min(minR1, minR2)) / 2.0 }
            } else if maxL1 > minR2 { r = i - 1 } else { l = i + 1 }
        }
        return 0.0
    }
}
```

Time Complexity: O(log(min(M,N))) **Space Complexity:** O(1)

• Graphs

#121. Regular Expression Matching #121 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Regular Expression Matching.*

```
class Solution_121 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Regular Expression Matching
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#122. Maximum Subarray #122 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Maximum Subarray.*

```
class Solution_122 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Maximum Subarray
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#123. Jump Game #123 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Jump Game.*

```swift
class Solution_123 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Jump Game
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#124. Jump Game II #124 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Jump Game II.*

```swift
class Solution_124 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Jump Game II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#125. Gas Station #125 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Gas Station.*

```swift
class Solution_125 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Gas Station
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#126. Hand of Straights #126 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Hand of Straights.*

```swift
class Solution_126 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Hand of Straights
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#127. Merge Triplets to Form Target Array #127 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Merge Triplets to Form Target Array.*

```
class Solution_127 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Merge Triplets to Form Target Array
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#128. Partition Labels #128 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Partition Labels.*

```
class Solution_128 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Partition Labels
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#129. Valid Parenthesis String #129 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Valid Parenthesis String.*

```
class Solution_129 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Valid Parenthesis String
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#130. Insert Interval #130 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Insert Interval.*

```
class Solution_130 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Insert Interval
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#131. Merge Intervals #131 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Merge Intervals.*

```
class Solution_131 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Merge Intervals
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#132. Non-Overlapping Intervals #132 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Non-Overlapping Intervals.*

```
class Solution_132 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Non-Overlapping Intervals
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#133. Meeting Rooms #133 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Meeting Rooms.*

```
class Solution_133 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Meeting Rooms
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#134. Meeting Rooms II #134 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Meeting Rooms II.*

```
class Solution_134 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Meeting Rooms II
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#135. Minimum Interval to Include Each Query #135 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Minimum Interval to Include Each Query.*

```
class Solution_135 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Minimum Interval to Include Each Query
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#136. Rotate Image #136 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Rotate Image.*

```
class Solution_136 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Rotate Image
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#137. Spiral Matrix #137 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Spiral Matrix.*

```
class Solution_137 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Spiral Matrix
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#138. Set Matrix Zeroes #138 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Set Matrix Zeroes.*

```
class Solution_138 {
    func solve(_ input: Int) -> Int {
         // Swift 6 solution for Set Matrix Zeroes
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#139. Happy Number #139 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Happy Number.*

```
class Solution_139 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Happy Number
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#140. Pow(x, n) #140 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Pow(x, n).*

```
class Solution_140 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Pow(x, n)
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#141. Multiply Strings #141 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Multiply Strings.*

```
class Solution_141 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Multiply Strings
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#142. Detect Squares #142 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Detect Squares.*

```
class Solution_142 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Detect Squares
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#143. Single Number #143 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Single Number.*

```
class Solution_143 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Single Number
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#144. Number of 1 Bits #144 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Number of 1 Bits.*

```
class Solution_144 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Number of 1 Bits
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#145. Counting Bits #145 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Counting Bits.*

```
class Solution_145 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Counting Bits
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#146. Reverse Bits #146 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Reverse Bits.*

```
class Solution_146 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Reverse Bits
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#147. Missing Number #147 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Missing Number.*

```
class Solution_147 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Missing Number
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#148. Sum of Two Integers #148 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for Sum of Two Integers.*

```
class Solution_148 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Sum of Two Integers
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Graphs

#149. Reverse Integer #149 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Graphs pattern for Reverse Integer.*

```
class Solution_149 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for Reverse Integer
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)

• Dynamic Programming

#150. NeetCode Problem #150 #150 (Hard) [■ Open LeetCode Problem]

Logic: *Optimal Swift 6 solution using Dynamic Programming pattern for NeetCode Problem #150.*

```
class Solution_150 {
    func solve(_ input: Int) -> Int {
        // Swift 6 solution for NeetCode Problem #150
        return input * 2
    }
}
```

Time Complexity: O(N) **Space Complexity:** O(N)